# Actuarial Modeling of Annuity-Based Life Insurance Products

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## Overview



- Implementation of actuarial logic for a life insurance product
- Annuity-based benefits with dynamic contract evolution
- Translation of actuarial specifications into production model
- Focus on:
  - numerical consistency
  - robustness
  - alignment with external models

# Mathematical Foundations



- Mortality framework:
  - $q_x$ – death probability, ${}_tp_x$ – survival probability from age $x$ to age $x + t$

**Net Single Premium (NSP):**

$$\text{NSP} = \sum_{t=1}^{T} v^t \cdot {}_tp_x \cdot B_t + \sum_{t=1}^{T} v^t \cdot {}_{t-1}p_x \cdot q_{x+t-1} \cdot D_t$$

- $v^t$ – discount factor
- survival component (annuity): ${}_tp_x \cdot B_t$
- death component: ${}_{t-1}p_x \cdot q_{x+t-1} \cdot D_t$

- key issue: correct interaction of survival vs. death cash flows

# Core Relationships



- Inverse relationships between:
  - premium
  - sum insured
  - annuity payout
- Implicit structure:

$$\text{premium} \leftrightarrow \text{sum insured} \leftrightarrow \text{annuity}$$

- Requires stable computational handling

## Pricing Model Issues



- Inconsistencies in mortality-based pricing formulas
- Net Single Premium did not fit death scenarios
- Mismatch between:
  - discrete mortality structure
  - provided formula assumptions
- Required model correction and validation

## Inverse Calculations



- Solving non-explicit relationships:
  - premium $\rightarrow$ benefit
  - benefit $\rightarrow$ premium
- Key challenges:
  - numerical stability
  - consistency across modes
  - sensitivity to input parameters

## Dynamic Contract Changes



- Mid-term contract modifications:
  - premium adjustments
  - sum insured changes
  - annuity changes
- Requires:
  - recalculation of dependent variables
  - preservation of internal consistency

## Reserve System



- Multiple reserve types:
  - net reserve
  - gross (Zillmer) reserve
  - surrender value
  - paid-up benefits
- Interdependent structure across time

## Implementation

- Integration into production environment
- Structured calculation pipeline
- Dependency management between components
- Removal of redundant calculations

## External Model Alignment

- Alignment with spreadsheet-based reference models
- Reverse engineering of calculation logic
- Reproduction of expected outputs
- Resolution of discrepancies

# What was actually solved



- Correction of inconsistent mortality-based pricing logic
- Stabilization of inverse actuarial calculations
- Design of a consistent recalculation framework for dynamic contracts
- Translation of actuarial specifications into reliable production code

*Core contribution: bridging actuarial mathematics and production systems under real-world constraints.*

## Outcome



- Robust actuarial model for pricing and reserves
- Support for dynamic contract changes
- Numerically stable implementation
- Consistency with external specifications